

Shahriyar Shawon

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Experience

Software Engineer Intern - SFG

Bala Cynwyd, PA

Susquehanna International Group | C#, .NET, OracleDB, MySQL, WPF

September 2023 – March 2024

- Developed an API to aggregate data from 15 databases to enhance accuracy for stock information query
- Redesigned web app to use central API to simplify data sourcing
- Introduced caching mechanism by front loading requests to reduce lookup speed of stock information by 95%
- Designed and implemented role based security access system to protect privileged data

Software Engineer Intern - MDF Metadata

Bala Cynwyd, PA

Susquehanna International Group | Python, Django, MySQL, PostgreSQL, Linux

September 2022 – March 2023

- Developed an alerting system using Python and PostgreSQL to ensure files were received correctly and on time, improving data accuracy and workflow efficiency
- Simplified the process of creating alerts by developing an command line interface, reducing setup time and increasing developer user productivity
- Enhanced the accuracy of a tool for identifying users of internal files by implementing multiple identification methods, leading to more reliable user data

Software Engineer Intern - CorpSys

Bala Cynwyd, PA

Susquehanna International Group | C#, SQL, .NET Framework/Core

September 2021 – August 2022

- Enriched an internal project management tool by gathering up-to-date information about developers, stakeholders, testing pipelines, environments, and code location, improving team collaboration and project tracking
- Developed the front-end for an API to allow general-purpose alerts to be created, enhancing user notification capabilities
- Reduced the number of bridge tables in MySQL by using Entity Framework ORM to make queries, leading to more efficient database operations and improved application performance

Projects

Dynamic Image Compression Engine | C, Python, RP2040

- Built firmware for 2 RP2040 chips to interface with FPGA, IO Modules, and Power Monitor Chips
- Implemented dynamic image format to store lossy and lossless blocks of data using QOI and PNG codecs
- Created desktop app to control camera settings on the fly and log power/data metrics from hardware
- Engineered pipeline to compare software based encoding performance to hardware based encoding

Chip 8 Emulator | Rust

- Modeled Chip 8 CPU as data structure to maintain internal state
- Implemented Fetch-Decode-Execute loop on Chip 8 ROMs
- Created disassembler to assist in emulator debugging
- Used SDL to visualize video memory contents

Drexel Course Information API | FastAPI, grequests, BeautifulSoup, Web Scraping, REST

- Built web scraper to parse information from 8200+ Drexel courses and store in in a SQLite Database
- Created API to be able to query for classes based on school, subject, credits, and name
- Enabled users to see the all the prerequisites for classes

Technical Skills

Languages: Python, C#, C, C++, Rust, Go, Bash/Shell Scripting, Java, MySQL, PostgreSQL, OracleDB, Javascript

Frameworks: FastAPI, Django, Flask, .NET Core/Framework, Entity Framework, JQuery

Tools: Docker, Git, Github, Gitlab CI/CD, Jira, Confluence

Platforms: Linux, Windows, Docker, RP2040, Raspberry Pi, Arduino

Education

Drexel University

Philadelphia, PA

Bachelor of Science in Computer Science

September 2020 – June 2025

Concentrations: Operating Systems and Computational Imaging